FAULT CURRENT STUDIO - ENGINEERING REPORT
BROWSER EXPORT - ENGINEERING REVIEW REQUIRED

===== SUMMARY =====
Project: Fault Current Studio advanced study
Method: multienvelope
Model status: DOCUMENTED_SCENARIO_ENVELOPE_SCREEN
Captured UTC: 2026-10-26T06:04:37.512Z
Warning: ENGINEERING SCREENING ONLY - independently verify actual manufacturer, utility, installation, equipment settings and standards edition before professional use.
Report data provenance: Entered study inputs, isolated calculated method outputs. No external OEM certification or auto-selected breaker/PPE.

===== STUDY INPUTS =====
Input path: studyBasis | Entered value: Illustrative two-case envelope, NOT certified utility maximum/minimum IEC 60909 operating conditions.
Input path: scenarios[0].id | Entered value: example_high
Input path: scenarios[0].sourceReference | Entered value: illustrative-source-conditions-A
Input path: scenarios[0].network.nominalLLV | Entered value: 13800
Input path: scenarios[0].network.voltageFactor | Entered value: 1.1
Input path: scenarios[0].network.frequencyHz | Entered value: 60
Input path: scenarios[0].network.faultType | Entered value: AG
Input path: scenarios[0].network.faultBus | Entered value: F
Input path: scenarios[0].network.buses[0] | Entered value: F
Input path: scenarios[0].network.buses[1] | Entered value: A
Input path: scenarios[0].network.buses[2] | Entered value: B
Input path: scenarios[0].network.branches[0].id | Entered value: line-A
Input path: scenarios[0].network.branches[0].from | Entered value: A
Input path: scenarios[0].network.branches[0].to | Entered value: F
Input path: scenarios[0].network.branches[0].positiveSequence.rOhm | Entered value: 0.06
Input path: scenarios[0].network.branches[0].positiveSequence.xOhm | Entered value: 0.12
Input path: scenarios[0].network.branches[0].negativeSequence.rOhm | Entered value: 0.07
Input path: scenarios[0].network.branches[0].negativeSequence.xOhm | Entered value: 0.14
Input path: scenarios[0].network.branches[0].zeroSequence.rOhm | Entered value: 0.23
Input path: scenarios[0].network.branches[0].zeroSequence.xOhm | Entered value: 0.44
Input path: scenarios[0].network.branches[1].id | Entered value: line-B
Input path: scenarios[0].network.branches[1].from | Entered value: B
Input path: scenarios[0].network.branches[1].to | Entered value: F
Input path: scenarios[0].network.branches[1].positiveSequence.rOhm | Entered value: 0.04
Input path: scenarios[0].network.branches[1].positiveSequence.xOhm | Entered value: 0.13
Input path: scenarios[0].network.branches[1].negativeSequence.rOhm | Entered value: 0.05
Input path: scenarios[0].network.branches[1].negativeSequence.xOhm | Entered value: 0.16
Input path: scenarios[0].network.branches[1].zeroSequence.rOhm | Entered value: 0.22
Input path: scenarios[0].network.branches[1].zeroSequence.xOhm | Entered value: 0.48
Input path: scenarios[0].network.sources[0].id | Entered value: grid-A
Input path: scenarios[0].network.sources[0].bus | Entered value: A
Input path: scenarios[0].network.sources[0].positiveSequence.rOhm | Entered value: 0.08
Input path: scenarios[0].network.sources[0].positiveSequence.xOhm | Entered value: 0.42
Input path: scenarios[0].network.sources[0].negativeSequence.rOhm | Entered value: 0.09
Input path: scenarios[0].network.sources[0].negativeSequence.xOhm | Entered value: 0.46
Input path: scenarios[0].network.sources[0].zeroSequence.rOhm | Entered value: 0.31
Input path: scenarios[0].network.sources[0].zeroSequence.xOhm | Entered value: 1.2
Input path: scenarios[0].network.sources[0].angleDeg | Entered value: 0
Input path: scenarios[0].network.sources[0].emScale | Entered value: 1
Input path: scenarios[0].network.sources[1].id | Entered value: source-B
Input path: scenarios[0].network.sources[1].bus | Entered value: B
Input path: scenarios[0].network.sources[1].positiveSequence.rOhm | Entered value: 0.11
Input path: scenarios[0].network.sources[1].positiveSequence.xOhm | Entered value: 0.6

Input path: scenarios[0].network.sources[1].negativeSequence.rOhm | Entered value: 0.13
Input path: scenarios[0].network.sources[1].negativeSequence.xOhm | Entered value: 0.67
Input path: scenarios[0].network.sources[1].zeroSequence.rOhm | Entered value: 0.33
Input path: scenarios[0].network.sources[1].zeroSequence.xOhm | Entered value: 1.34
Input path: scenarios[0].network.sources[1].angleDeg | Entered value: 0
Input path: scenarios[0].network.sources[1].emScale | Entered value: 1
Input path: scenarios[1].id | Entered value: example_low
Input path: scenarios[1].sourceReference | Entered value: illustrative-source-conditions-B
Input path: scenarios[1].network.nominalLLV | Entered value: 13800
Input path: scenarios[1].network.voltageFactor | Entered value: 0.95
Input path: scenarios[1].network.frequencyHz | Entered value: 60
Input path: scenarios[1].network.faultType | Entered value: AG
Input path: scenarios[1].network.faultBus | Entered value: F
Input path: scenarios[1].network.buses[0] | Entered value: F
Input path: scenarios[1].network.buses[1] | Entered value: A
Input path: scenarios[1].network.buses[2] | Entered value: B
Input path: scenarios[1].network.branches[0].id | Entered value: line-A
Input path: scenarios[1].network.branches[0].from | Entered value: A
Input path: scenarios[1].network.branches[0].to | Entered value: F
Input path: scenarios[1].network.branches[0].positiveSequence.rOhm | Entered value: 0.06
Input path: scenarios[1].network.branches[0].positiveSequence.xOhm | Entered value: 0.12
Input path: scenarios[1].network.branches[0].negativeSequence.rOhm | Entered value: 0.07
Input path: scenarios[1].network.branches[0].negativeSequence.xOhm | Entered value: 0.14
Input path: scenarios[1].network.branches[0].zeroSequence.rOhm | Entered value: 0.23
Input path: scenarios[1].network.branches[0].zeroSequence.xOhm | Entered value: 0.44
Input path: scenarios[1].network.branches[1].id | Entered value: line-B
Input path: scenarios[1].network.branches[1].from | Entered value: B
Input path: scenarios[1].network.branches[1].to | Entered value: F
Input path: scenarios[1].network.branches[1].positiveSequence.rOhm | Entered value: 0.04
Input path: scenarios[1].network.branches[1].positiveSequence.xOhm | Entered value: 0.13
Input path: scenarios[1].network.branches[1].negativeSequence.rOhm | Entered value: 0.05
Input path: scenarios[1].network.branches[1].negativeSequence.xOhm | Entered value: 0.16
Input path: scenarios[1].network.branches[1].zeroSequence.rOhm | Entered value: 0.22
Input path: scenarios[1].network.branches[1].zeroSequence.xOhm | Entered value: 0.48
Input path: scenarios[1].network.sources[0].id | Entered value: grid-A
Input path: scenarios[1].network.sources[0].bus | Entered value: A
Input path: scenarios[1].network.sources[0].positiveSequence.rOhm | Entered value: 0.08
Input path: scenarios[1].network.sources[0].positiveSequence.xOhm | Entered value: 0.42
Input path: scenarios[1].network.sources[0].negativeSequence.rOhm | Entered value: 0.09
Input path: scenarios[1].network.sources[0].negativeSequence.xOhm | Entered value: 0.46
Input path: scenarios[1].network.sources[0].zeroSequence.rOhm | Entered value: 0.31
Input path: scenarios[1].network.sources[0].zeroSequence.xOhm | Entered value: 1.2
Input path: scenarios[1].network.sources[0].angleDeg | Entered value: 0
Input path: scenarios[1].network.sources[0].emScale | Entered value: 1
Input path: scenarios[1].network.sources[1].id | Entered value: source-B
Input path: scenarios[1].network.sources[1].bus | Entered value: B
Input path: scenarios[1].network.sources[1].positiveSequence.rOhm | Entered value: 0.11
Input path: scenarios[1].network.sources[1].positiveSequence.xOhm | Entered value: 0.6
Input path: scenarios[1].network.sources[1].negativeSequence.rOhm | Entered value: 0.13
Input path: scenarios[1].network.sources[1].negativeSequence.xOhm | Entered value: 0.67
Input path: scenarios[1].network.sources[1].zeroSequence.rOhm | Entered value: 0.33
Input path: scenarios[1].network.sources[1].zeroSequence.xOhm | Entered value: 1.34
Input path: scenarios[1].network.sources[1].angleDeg | Entered value: 0
Input path: scenarios[1].network.sources[1].emScale | Entered value: 0.87

===== EQUIPMENT DUTY =====
Calculated output path: status | Value: DOCUMENTED_SCENARIO_ENVELOPE_SCREEN
Calculated output path: studyBasis | Value: Illustrative two-case envelope, NOT certified utility maximum/minimum IEC 60909 operating conditions.

Calculated output path: scenarioCount | Value: 2
Calculated output path: maximumScenarioId | Value: example_high
Calculated output path: minimumScenarioId | Value: example_low
Calculated output path: maximumPhaseRmsA | Value: 16620.557944533848
Calculated output path: minimumPhaseRmsA | Value: 13555.382764706039
Calculated output path: ratioHighToLow | Value: 1.2261223628305475
Calculated output path: scenarios[0].id | Value: example_high
Calculated output path: scenarios[0].sourceReference | Value: illustrative-source-conditions-A
Calculated output path: scenarios[0].maximumPhaseRmsA | Value: 16620.557944533848
Calculated output path: scenarios[0].groundResidualA | Value: 16620.557944533844
Calculated output path: scenarios[0].faultType | Value: AG
Calculated output path: scenarios[0].networkResult.status | Value: BOUNDED_MULTI_BUS_SEQUENCE_FAULT
Calculated output path: scenarios[0].networkResult.scope | Value: One voltage base, explicitly entered positive/negative/zero-sequence branch and source impedances, independent fixed positive-sequence source EMFs; four metallic fault configurations. No IEC 60909 corrections, transformer vector shifts, time-dependent generator/IBR currents or making/breaking duties.
Calculated output path: scenarios[0].networkResult.faultType | Value: AG
Calculated output path: scenarios[0].networkResult.nominalLLV | Value: 13800
Calculated output path: scenarios[0].networkResult.voltageFactor | Value: 1.1
Calculated output path: scenarios[0].networkResult.frequencyHz | Value: 60
Calculated output path: scenarios[0].networkResult.faultBus | Value: F
Calculated output path: scenarios[0].networkResult.faultImpedanceOhm.rOhm | Value: 0
Calculated output path: scenarios[0].networkResult.faultImpedanceOhm.xOhm | Value: 0
Calculated output path: scenarios[0].networkResult.sequenceTheveninOhm[0].real | Value: 0.2729322097008292
Calculated output path: scenarios[0].networkResult.sequenceTheveninOhm[0].imag | Value: 0.8628023435184697
Calculated output path: scenarios[0].networkResult.sequenceTheveninOhm[0].magnitude | Value: 0.9049419180660937
Calculated output path: scenarios[0].networkResult.sequenceTheveninOhm[0].angleDegrees | Value: 72.44615709955977
Calculated output path: scenarios[0].networkResult.sequenceTheveninOhm[1].real | Value: 0.07332704773129058
Calculated output path: scenarios[0].networkResult.sequenceTheveninOhm[1].imag | Value: 0.3106022392457279
Calculated output path: scenarios[0].networkResult.sequenceTheveninOhm[1].magnitude | Value: 0.31914041886518757
Calculated output path: scenarios[0].networkResult.sequenceTheveninOhm[1].angleDegrees | Value: 76.7168095145296
Calculated output path: scenarios[0].networkResult.sequenceTheveninOhm[2].real | Value: 0.08554316130525336
Calculated output path: scenarios[0].networkResult.sequenceTheveninOhm[2].imag | Value: 0.34845082156908147
Calculated output path: scenarios[0].networkResult.sequenceTheveninOhm[2].magnitude | Value: 0.3587974463374349
Calculated output path: scenarios[0].networkResult.sequenceTheveninOhm[2].angleDegrees | Value: 76.20691177731727
Calculated output path: scenarios[0].networkResult.prefaultPositiveFaultBusV.real | Value: 8764.177086298525
Calculated output path: scenarios[0].networkResult.prefaultPositiveFaultBusV.imag | Value: -1.0694152322308772e-12
Calculated output path: scenarios[0].networkResult.prefaultPositiveFaultBusV.magnitude | Value: 8764.177086298525
Calculated output path: scenarios[0].networkResult.prefaultPositiveFaultBusV.angleDegrees | Value: -6.991298641103816e-15
Calculated output path: scenarios[0].networkResult.sequenceFaultCurrentA[0].real | Value: 1512.2465924481921
Calculated output path: scenarios[0].networkResult.sequenceFaultCurrentA[0].imag | Value: -5329.800273308838

Calculated output path: scenarios[0].networkResult.sequenceFaultCurrentA[0].magnitude | Value:
5540.185981511282
Calculated output path: scenarios[0].networkResult.sequenceFaultCurrentA[0].angleDegrees | Value:
-74.15954736119843
Calculated output path: scenarios[0].networkResult.sequenceFaultCurrentA[1].real | Value:
1512.2465924481921
Calculated output path: scenarios[0].networkResult.sequenceFaultCurrentA[1].imag | Value:
-5329.800273308838
Calculated output path: scenarios[0].networkResult.sequenceFaultCurrentA[1].magnitude | Value:
5540.185981511282
Calculated output path: scenarios[0].networkResult.sequenceFaultCurrentA[1].angleDegrees | Value:
-74.15954736119843
Calculated output path: scenarios[0].networkResult.sequenceFaultCurrentA[2].real | Value:
1512.2465924481921
Calculated output path: scenarios[0].networkResult.sequenceFaultCurrentA[2].imag | Value:
-5329.800273308838
Calculated output path: scenarios[0].networkResult.sequenceFaultCurrentA[2].magnitude | Value:
5540.185981511282
Calculated output path: scenarios[0].networkResult.sequenceFaultCurrentA[2].angleDegrees | Value:
-74.15954736119843
Calculated output path: scenarios[0].networkResult.phaseFaultCurrentA[0].real | Value:
4536.739777344576
Calculated output path: scenarios[0].networkResult.phaseFaultCurrentA[0].imag | Value:
-15989.400819926514
Calculated output path: scenarios[0].networkResult.phaseFaultCurrentA[0].magnitude | Value:
16620.557944533848
Calculated output path: scenarios[0].networkResult.phaseFaultCurrentA[0].angleDegrees | Value:
-74.15954736119843
Calculated output path: scenarios[0].networkResult.phaseFaultCurrentA[1].real | Value:
4.547473508864641e-13
Calculated output path: scenarios[0].networkResult.phaseFaultCurrentA[1].imag | Value: 0
Calculated output path: scenarios[0].networkResult.phaseFaultCurrentA[1].magnitude | Value:
4.547473508864641e-13
Calculated output path: scenarios[0].networkResult.phaseFaultCurrentA[1].angleDegrees | Value: 0
Calculated output path: scenarios[0].networkResult.phaseFaultCurrentA[2].real | Value: 0
Calculated output path: scenarios[0].networkResult.phaseFaultCurrentA[2].imag | Value: 0
Calculated output path: scenarios[0].networkResult.phaseFaultCurrentA[2].magnitude | Value: 0
Calculated output path: scenarios[0].networkResult.phaseFaultCurrentA[2].angleDegrees | Value: 0
Calculated output path: scenarios[0].networkResult.groundResidualA | Value: 16620.557944533844
Calculated output path: scenarios[0].networkResult.maximumPhaseRmsA | Value: 16620.557944533848
Calculated output path: scenarios[0].networkResult.sourceContributions[0].id | Value: grid-A
Calculated output path: scenarios[0].networkResult.sourceContributions[0].bus | Value: A
Calculated output path: scenarios[0].networkResult.sourceContributions[0].phaseCurrentsA[0].real |
Value: 2691.562397669685
Calculated output path: scenarios[0].networkResult.sourceContributions[0].phaseCurrentsA[0].imag |
Value: -8875.047550729882
Calculated output path: scenarios[0].networkResult.sourceContributions[0].phaseCurrentsA[0].magnitu
de | Value: 9274.210325858794
Calculated output path: scenarios[0].networkResult.sourceContributions[0].phaseCurrentsA[0].angleDe
grees | Value: -73.12886548949481
Calculated output path: scenarios[0].networkResult.sourceContributions[0].phaseCurrentsA[1].real |
Value: -80.07152598509356
Calculated output path: scenarios[0].networkResult.sourceContributions[0].phaseCurrentsA[1].imag |
Value: 264.44816588912727
Calculated output path: scenarios[0].networkResult.sourceContributions[0].phaseCurrentsA[1].magnitu
de | Value: 276.304689999473
Calculated output path: scenarios[0].networkResult.sourceContributions[0].phaseCurrentsA[1].angleDe
grees | Value: 106.8456418478821
Calculated output path: scenarios[0].networkResult.sourceContributions[0].phaseCurrentsA[2].real |

Value: -135.7143107704046
Calculated output path: scenarios[0].networkResult.sourceContributions[0].phaseCurrentsA[2].imag | Value: 258.7728188264974
Calculated output path: scenarios[0].networkResult.sourceContributions[0].phaseCurrentsA[2].magnitude | Value: 292.20155015211196
Calculated output path: scenarios[0].networkResult.sourceContributions[0].phaseCurrentsA[2].angleDegrees | Value: 117.67492081629787
Calculated output path: scenarios[0].networkResult.sourceContributions[0].sequenceCurrentsA[0].real | Value: 825.2588536380623
Calculated output path: scenarios[0].networkResult.sourceContributions[0].sequenceCurrentsA[0].imag | Value: -2783.942188671419
Calculated output path: scenarios[0].networkResult.sourceContributions[0].sequenceCurrentsA[0].magnitude | Value: 2903.684949400093
Calculated output path: scenarios[0].networkResult.sourceContributions[0].sequenceCurrentsA[0].angleDegrees | Value: -73.48831515028421
Calculated output path: scenarios[0].networkResult.sourceContributions[0].sequenceCurrentsA[1].real | Value: 931.5134404386342
Calculated output path: scenarios[0].networkResult.sourceContributions[0].sequenceCurrentsA[1].imag | Value: -3029.4899926421012
Calculated output path: scenarios[0].networkResult.sourceContributions[0].sequenceCurrentsA[1].magnitude | Value: 3169.467921471435
Calculated output path: scenarios[0].networkResult.sourceContributions[0].sequenceCurrentsA[1].angleDegrees | Value: -72.90828276612866
Calculated output path: scenarios[0].networkResult.sourceContributions[0].sequenceCurrentsA[2].real | Value: 934.7901035929883
Calculated output path: scenarios[0].networkResult.sourceContributions[0].sequenceCurrentsA[2].imag | Value: -3061.6153694163613
Calculated output path: scenarios[0].networkResult.sourceContributions[0].sequenceCurrentsA[2].magnitude | Value: 3201.1437343583734
Calculated output path: scenarios[0].networkResult.sourceContributions[0].sequenceCurrentsA[2].angleDegrees | Value: -73.02121840574361
Calculated output path: scenarios[0].networkResult.sourceContributions[1].id | Value: source-B
Calculated output path: scenarios[0].networkResult.sourceContributions[1].bus | Value: B
Calculated output path: scenarios[0].networkResult.sourceContributions[1].phaseCurrentsA[0].real | Value: 1845.177379674894
Calculated output path: scenarios[0].networkResult.sourceContributions[1].phaseCurrentsA[0].imag | Value: -7114.353269196625
Calculated output path: scenarios[0].networkResult.sourceContributions[1].phaseCurrentsA[0].magnitude | Value: 7349.741628206573
Calculated output path: scenarios[0].networkResult.sourceContributions[1].phaseCurrentsA[0].angleDegrees | Value: -75.4601462566015
Calculated output path: scenarios[0].networkResult.sourceContributions[1].phaseCurrentsA[1].real | Value: 80.07152598510766
Calculated output path: scenarios[0].networkResult.sourceContributions[1].phaseCurrentsA[1].imag | Value: -264.44816588913363
Calculated output path: scenarios[0].networkResult.sourceContributions[1].phaseCurrentsA[1].magnitude | Value: 276.3046899994832
Calculated output path: scenarios[0].networkResult.sourceContributions[1].phaseCurrentsA[1].angleDegrees | Value: -73.15435815211549
Calculated output path: scenarios[0].networkResult.sourceContributions[1].phaseCurrentsA[2].real | Value: 135.7143107703905
Calculated output path: scenarios[0].networkResult.sourceContributions[1].phaseCurrentsA[2].imag | Value: -258.77281882651005
Calculated output path: scenarios[0].networkResult.sourceContributions[1].phaseCurrentsA[2].magnitude | Value: 292.2015501521166
Calculated output path: scenarios[0].networkResult.sourceContributions[1].phaseCurrentsA[2].angleDegrees | Value: -62.32507918370572
Calculated output path: scenarios[0].networkResult.sourceContributions[1].sequenceCurrentsA[0].real | Value: 686.9877388101307

Calculated output path: scenarios[0].networkResult.sourceContributions[1].sequenceCurrentsA[0].imag
| Value: -2545.8580846374225
Calculated output path: scenarios[0].networkResult.sourceContributions[1].sequenceCurrentsA[0].magn
itude | Value: 2636.9197068528993
Calculated output path: scenarios[0].networkResult.sourceContributions[1].sequenceCurrentsA[0].angl
eDegrees | Value: -74.8986886937207
Calculated output path: scenarios[0].networkResult.sourceContributions[1].sequenceCurrentsA[1].real
| Value: 580.7331520095568
Calculated output path: scenarios[0].networkResult.sourceContributions[1].sequenceCurrentsA[1].imag
| Value: -2300.3102806667225
Calculated output path: scenarios[0].networkResult.sourceContributions[1].sequenceCurrentsA[1].magn
itude | Value: 2372.4835892338583
Calculated output path: scenarios[0].networkResult.sourceContributions[1].sequenceCurrentsA[1].angl
eDegrees | Value: -75.83125142278031
Calculated output path: scenarios[0].networkResult.sourceContributions[1].sequenceCurrentsA[2].real
| Value: 577.4564888552064
Calculated output path: scenarios[0].networkResult.sourceContributions[1].sequenceCurrentsA[2].imag
| Value: -2268.184903892479
Calculated output path: scenarios[0].networkResult.sourceContributions[1].sequenceCurrentsA[2].magn
itude | Value: 2340.5381335852485
Calculated output path: scenarios[0].networkResult.sourceContributions[1].sequenceCurrentsA[2].angl
eDegrees | Value: -75.71652407446481
Calculated output path: scenarios[0].networkResult.branchCurrents[0].id | Value: line-A
Calculated output path: scenarios[0].networkResult.branchCurrents[0].from | Value: A
Calculated output path: scenarios[0].networkResult.branchCurrents[0].to | Value: F
Calculated output path: scenarios[0].networkResult.branchCurrents[0].phaseCurrentsA[0].real |
Value: 2691.5623976696897
Calculated output path: scenarios[0].networkResult.branchCurrents[0].phaseCurrentsA[0].imag |
Value: -8875.047550729887
Calculated output path: scenarios[0].networkResult.branchCurrents[0].phaseCurrentsA[0].magnitude |
Value: 9274.2103258588
Calculated output path: scenarios[0].networkResult.branchCurrents[0].phaseCurrentsA[0].angleDegrees
| Value: -73.12886548949479
Calculated output path: scenarios[0].networkResult.branchCurrents[0].phaseCurrentsA[1].real |
Value: -80.07152598510356
Calculated output path: scenarios[0].networkResult.branchCurrents[0].phaseCurrentsA[1].imag |
Value: 264.44816588912636
Calculated output path: scenarios[0].networkResult.branchCurrents[0].phaseCurrentsA[1].magnitude |
Value: 276.30468999947504
Calculated output path: scenarios[0].networkResult.branchCurrents[0].phaseCurrentsA[1].angleDegrees
| Value: 106.84564184788417
Calculated output path: scenarios[0].networkResult.branchCurrents[0].phaseCurrentsA[2].real |
Value: -135.71431077039915
Calculated output path: scenarios[0].networkResult.branchCurrents[0].phaseCurrentsA[2].imag |
Value: 258.772818826507
Calculated output path: scenarios[0].networkResult.branchCurrents[0].phaseCurrentsA[2].magnitude |
Value: 292.20155015211793
Calculated output path: scenarios[0].networkResult.branchCurrents[0].phaseCurrentsA[2].angleDegrees
| Value: 117.67492081629607
Calculated output path: scenarios[0].networkResult.branchCurrents[0].sequenceCurrentsA[0].real |
Value: 825.2588536380622
Calculated output path: scenarios[0].networkResult.branchCurrents[0].sequenceCurrentsA[0].imag |
Value: -2783.942188671418
Calculated output path: scenarios[0].networkResult.branchCurrents[0].sequenceCurrentsA[0].magnitude
| Value: 2903.6849494000917
Calculated output path: scenarios[0].networkResult.branchCurrents[0].sequenceCurrentsA[0].angleDegr
ees | Value: -73.48831515028421
Calculated output path: scenarios[0].networkResult.branchCurrents[0].sequenceCurrentsA[1].real |
Value: 931.5134404386398

Calculated output path: scenarios[0].networkResult.branchCurrents[0].sequenceCurrentsA[1].imag | Value: -3029.4899926421094
Calculated output path: scenarios[0].networkResult.branchCurrents[0].sequenceCurrentsA[1].magnitude | Value: 3169.4679214714442
Calculated output path: scenarios[0].networkResult.branchCurrents[0].sequenceCurrentsA[1].angleDegrees | Value: -72.90828276612862
Calculated output path: scenarios[0].networkResult.branchCurrents[0].sequenceCurrentsA[2].real | Value: 934.7901035929874
Calculated output path: scenarios[0].networkResult.branchCurrents[0].sequenceCurrentsA[2].imag | Value: -3061.61536941636
Calculated output path: scenarios[0].networkResult.branchCurrents[0].sequenceCurrentsA[2].magnitude | Value: 3201.143734358372
Calculated output path: scenarios[0].networkResult.branchCurrents[0].sequenceCurrentsA[2].angleDegrees | Value: -73.02121840574361
Calculated output path: scenarios[0].networkResult.branchCurrents[1].id | Value: line-B
Calculated output path: scenarios[0].networkResult.branchCurrents[1].from | Value: B
Calculated output path: scenarios[0].networkResult.branchCurrents[1].to | Value: F
Calculated output path: scenarios[0].networkResult.branchCurrents[1].phaseCurrentsA[0].real | Value: 1845.1773796749053
Calculated output path: scenarios[0].networkResult.branchCurrents[1].phaseCurrentsA[0].imag | Value: -7114.353269196645
Calculated output path: scenarios[0].networkResult.branchCurrents[1].phaseCurrentsA[0].magnitude | Value: 7349.741628206596
Calculated output path: scenarios[0].networkResult.branchCurrents[1].phaseCurrentsA[0].angleDegrees | Value: -75.46014625660146
Calculated output path: scenarios[0].networkResult.branchCurrents[1].phaseCurrentsA[1].real | Value: 80.07152598508105
Calculated output path: scenarios[0].networkResult.branchCurrents[1].phaseCurrentsA[1].imag | Value: -264.44816588913136
Calculated output path: scenarios[0].networkResult.branchCurrents[1].phaseCurrentsA[1].magnitude | Value: 276.3046899994733
Calculated output path: scenarios[0].networkResult.branchCurrents[1].phaseCurrentsA[1].angleDegrees | Value: -73.15435815212062
Calculated output path: scenarios[0].networkResult.branchCurrents[1].phaseCurrentsA[2].real | Value: 135.71431077040643
Calculated output path: scenarios[0].networkResult.branchCurrents[1].phaseCurrentsA[2].imag | Value: -258.77281882648913
Calculated output path: scenarios[0].networkResult.branchCurrents[1].phaseCurrentsA[2].magnitude | Value: 292.2015501521055
Calculated output path: scenarios[0].networkResult.branchCurrents[1].phaseCurrentsA[2].angleDegrees | Value: -62.32507918370105
Calculated output path: scenarios[0].networkResult.branchCurrents[1].sequenceCurrentsA[0].real | Value: 686.987738810131
Calculated output path: scenarios[0].networkResult.branchCurrents[1].sequenceCurrentsA[0].imag | Value: -2545.8580846374216
Calculated output path: scenarios[0].networkResult.branchCurrents[1].sequenceCurrentsA[0].magnitude | Value: 2636.9197068528983
Calculated output path: scenarios[0].networkResult.branchCurrents[1].sequenceCurrentsA[0].angleDegrees | Value: -74.89868869372069
Calculated output path: scenarios[0].networkResult.branchCurrents[1].sequenceCurrentsA[1].real | Value: 580.7331520095677
Calculated output path: scenarios[0].networkResult.branchCurrents[1].sequenceCurrentsA[1].imag | Value: -2300.3102806667453
Calculated output path: scenarios[0].networkResult.branchCurrents[1].sequenceCurrentsA[1].magnitude | Value: 2372.483589233883
Calculated output path: scenarios[0].networkResult.branchCurrents[1].sequenceCurrentsA[1].angleDegrees | Value: -75.8312514227802
Calculated output path: scenarios[0].networkResult.branchCurrents[1].sequenceCurrentsA[2].real | Value: 577.4564888552067

Calculated output path: scenarios[0].networkResult.branchCurrents[1].sequenceCurrentsA[2].imag | Value: -2268.1849038924775
Calculated output path: scenarios[0].networkResult.branchCurrents[1].sequenceCurrentsA[2].magnitude | Value: 2340.5381335852467
Calculated output path: scenarios[0].networkResult.branchCurrents[1].sequenceCurrentsA[2].angleDegrees | Value: -75.71652407446481
Calculated output path: scenarios[0].networkResult.busVoltages[0].bus | Value: F
Calculated output path: scenarios[0].networkResult.busVoltages[0].phaseVoltagesV[0].real | Value: 1.5916157281026244e-12
Calculated output path: scenarios[0].networkResult.busVoltages[0].phaseVoltagesV[0].imag | Value: 4.689582056016661e-13
Calculated output path: scenarios[0].networkResult.busVoltages[0].phaseVoltagesV[0].magnitude | Value: 1.6592656280851425e-12
Calculated output path: scenarios[0].networkResult.busVoltages[0].phaseVoltagesV[0].angleDegrees | Value: 16.417232830434777
Calculated output path: scenarios[0].networkResult.busVoltages[0].phaseVoltagesV[1].real | Value: -7523.775721922144
Calculated output path: scenarios[0].networkResult.busVoltages[0].phaseVoltagesV[1].imag | Value: -7555.841674042445
Calculated output path: scenarios[0].networkResult.busVoltages[0].phaseVoltagesV[1].magnitude | Value: 10662.923825901677
Calculated output path: scenarios[0].networkResult.busVoltages[0].phaseVoltagesV[1].angleDegrees | Value: -134.87816395435416
Calculated output path: scenarios[0].networkResult.busVoltages[0].phaseVoltagesV[2].real | Value: -7510.139189234899
Calculated output path: scenarios[0].networkResult.busVoltages[0].phaseVoltagesV[2].imag | Value: 8005.554459790879
Calculated output path: scenarios[0].networkResult.busVoltages[0].phaseVoltagesV[2].magnitude | Value: 10976.843483003639
Calculated output path: scenarios[0].networkResult.busVoltages[0].phaseVoltagesV[2].angleDegrees | Value: 133.17117015805255
Calculated output path: scenarios[0].networkResult.busVoltages[0].sequenceVoltagesV[0].real | Value: -5011.3049703856805
Calculated output path: scenarios[0].networkResult.busVoltages[0].sequenceVoltagesV[0].imag | Value: 149.90426191614415
Calculated output path: scenarios[0].networkResult.busVoltages[0].sequenceVoltagesV[0].magnitude | Value: 5013.546528551706
Calculated output path: scenarios[0].networkResult.busVoltages[0].sequenceVoltagesV[0].angleDegrees | Value: 178.28660973836136
Calculated output path: scenarios[0].networkResult.busVoltages[0].sequenceVoltagesV[1].real | Value: 6997.840608610377
Calculated output path: scenarios[0].networkResult.busVoltages[0].sequenceVoltagesV[1].imag | Value: -78.8886588669685
Calculated output path: scenarios[0].networkResult.busVoltages[0].sequenceVoltagesV[1].magnitude | Value: 6998.285261691915
Calculated output path: scenarios[0].networkResult.busVoltages[0].sequenceVoltagesV[1].angleDegrees | Value: -0.6458843512941284
Calculated output path: scenarios[0].networkResult.busVoltages[0].sequenceVoltagesV[2].real | Value: -1986.5356382246948
Calculated output path: scenarios[0].networkResult.busVoltages[0].sequenceVoltagesV[2].imag | Value: -71.01560304917518
Calculated output path: scenarios[0].networkResult.busVoltages[0].sequenceVoltagesV[2].magnitude | Value: 1987.804582400703
Calculated output path: scenarios[0].networkResult.busVoltages[0].sequenceVoltagesV[2].angleDegrees | Value: -177.95263558388115
Calculated output path: scenarios[0].networkResult.busVoltages[1].bus | Value: A
Calculated output path: scenarios[0].networkResult.busVoltages[1].phaseVoltagesV[0].real | Value: 2328.235163865351
Calculated output path: scenarios[0].networkResult.busVoltages[1].phaseVoltagesV[0].imag | Value:

-430.62305585569516
Calculated output path: scenarios[0].networkResult.busVoltages[1].phaseVoltagesV[0].magnitude |
Value: 2367.723631358528
Calculated output path: scenarios[0].networkResult.busVoltages[1].phaseVoltagesV[0].angleDegrees |
Value: -10.478827958056545
Calculated output path: scenarios[0].networkResult.busVoltages[1].phaseVoltagesV[1].real | Value:
-6554.12506477979
Calculated output path: scenarios[0].networkResult.busVoltages[1].phaseVoltagesV[1].imag | Value:
-7691.686276806313
Calculated output path: scenarios[0].networkResult.busVoltages[1].phaseVoltagesV[1].magnitude |
Value: 10105.37446835026
Calculated output path: scenarios[0].networkResult.busVoltages[1].phaseVoltagesV[1].angleDegrees |
Value: -130.43448596658666
Calculated output path: scenarios[0].networkResult.busVoltages[1].phaseVoltagesV[2].real | Value:
-6563.792712186067
Calculated output path: scenarios[0].networkResult.busVoltages[1].phaseVoltagesV[2].imag | Value:
7740.443695029404
Calculated output path: scenarios[0].networkResult.busVoltages[1].phaseVoltagesV[2].magnitude |
Value: 10148.785314729412
Calculated output path: scenarios[0].networkResult.busVoltages[1].phaseVoltagesV[2].angleDegrees |
Value: 130.29749874525513
Calculated output path: scenarios[0].networkResult.busVoltages[1].sequenceVoltagesV[0].real |
Value: -3596.5608710335023
Calculated output path: scenarios[0].networkResult.busVoltages[1].sequenceVoltagesV[0].imag |
Value: -127.28854587753472
Calculated output path: scenarios[0].networkResult.busVoltages[1].sequenceVoltagesV[0].magnitude |
Value: 3598.8126476604593
Calculated output path: scenarios[0].networkResult.busVoltages[1].sequenceVoltagesV[0].angleDegrees
| Value: -177.97304871060717
Calculated output path: scenarios[0].networkResult.busVoltages[1].sequenceVoltagesV[1].real |
Value: 7417.270214153748
Calculated output path: scenarios[0].networkResult.busVoltages[1].sequenceVoltagesV[1].imag |
Value: -148.87644557285827
Calculated output path: scenarios[0].networkResult.busVoltages[1].sequenceVoltagesV[1].magnitude |
Value: 7418.764157581692
Calculated output path: scenarios[0].networkResult.busVoltages[1].sequenceVoltagesV[1].angleDegrees
| Value: -1.14986329757395
Calculated output path: scenarios[0].networkResult.busVoltages[1].sequenceVoltagesV[2].real |
Value: -1492.4741792548953
Calculated output path: scenarios[0].networkResult.busVoltages[1].sequenceVoltagesV[2].imag |
Value: -154.45806440530214
Calculated output path: scenarios[0].networkResult.busVoltages[1].sequenceVoltagesV[2].magnitude |
Value: 1500.4454236667211
Calculated output path: scenarios[0].networkResult.busVoltages[1].sequenceVoltagesV[2].angleDegrees
| Value: -174.09142098368295
Calculated output path: scenarios[0].networkResult.busVoltages[2].bus | Value: B
Calculated output path: scenarios[0].networkResult.busVoltages[2].phaseVoltagesV[0].real | Value:
2087.201254796809
Calculated output path: scenarios[0].networkResult.busVoltages[2].phaseVoltagesV[0].imag | Value:
-267.8679724345863
Calculated output path: scenarios[0].networkResult.busVoltages[2].phaseVoltagesV[0].magnitude |
Value: 2104.3199207063526
Calculated output path: scenarios[0].networkResult.busVoltages[2].phaseVoltagesV[0].angleDegrees |
Value: -7.313269217975994
Calculated output path: scenarios[0].networkResult.busVoltages[2].phaseVoltagesV[1].real | Value:
-6503.7562349062455
Calculated output path: scenarios[0].networkResult.busVoltages[2].phaseVoltagesV[1].imag | Value:
-7707.209879457681
Calculated output path: scenarios[0].networkResult.busVoltages[2].phaseVoltagesV[1].magnitude |

Value: 10084.638282511274
Calculated output path: scenarios[0].networkResult.busVoltages[2].phaseVoltagesV[1].angleDegrees |
Value: -130.1594643542308
Calculated output path: scenarios[0].networkResult.busVoltages[2].phaseVoltagesV[2].real | Value:
-6497.912381555031
Calculated output path: scenarios[0].networkResult.busVoltages[2].phaseVoltagesV[2].imag | Value:
7733.78664566659
Calculated output path: scenarios[0].networkResult.busVoltages[2].phaseVoltagesV[2].magnitude |
Value: 10101.20394799833
Calculated output path: scenarios[0].networkResult.busVoltages[2].phaseVoltagesV[2].angleDegrees |
Value: 130.03691132992225
Calculated output path: scenarios[0].networkResult.busVoltages[2].sequenceVoltagesV[0].real |
Value: -3638.1557872214894
Calculated output path: scenarios[0].networkResult.busVoltages[2].sequenceVoltagesV[0].imag |
Value: -80.43040207522574
Calculated output path: scenarios[0].networkResult.busVoltages[2].sequenceVoltagesV[0].magnitude |
Value: 3639.044734771915
Calculated output path: scenarios[0].networkResult.busVoltages[2].sequenceVoltagesV[0].angleDegrees
| Value: -178.73354185037942
Calculated output path: scenarios[0].networkResult.busVoltages[2].sequenceVoltagesV[1].real |
Value: 7320.1102711774365
Calculated output path: scenarios[0].networkResult.busVoltages[2].sequenceVoltagesV[1].imag |
Value: -95.40576033239454
Calculated output path: scenarios[0].networkResult.busVoltages[2].sequenceVoltagesV[1].magnitude |
Value: 7320.731974420454
Calculated output path: scenarios[0].networkResult.busVoltages[2].sequenceVoltagesV[1].angleDegrees
| Value: -0.7467152428685605
Calculated output path: scenarios[0].networkResult.busVoltages[2].sequenceVoltagesV[2].real |
Value: -1594.753229159138
Calculated output path: scenarios[0].networkResult.busVoltages[2].sequenceVoltagesV[2].imag |
Value: -92.03181002696601
Calculated output path: scenarios[0].networkResult.busVoltages[2].sequenceVoltagesV[2].magnitude |
Value: 1597.4065593862879
Calculated output path: scenarios[0].networkResult.busVoltages[2].sequenceVoltagesV[2].angleDegrees
| Value: -176.6971740846384
Calculated output path: scenarios[0].networkResult.kclRelativeResidual | Value:
2.634614910300192e-15
Calculated output path: scenarios[0].networkResult.nodalRelativeResidual | Value:
1.1443916996305582e-15
Calculated output path: scenarios[0].networkResult.limitations[0] | Value: Explicit
positive/negative/zero-sequence equivalent impedances; no transformer winding shift or
zero-sequence grounding topology inference
Calculated output path: scenarios[0].networkResult.limitations[1] | Value: Multiple documented
fixed positive-sequence Thevenin EMFs, one voltage level only
Calculated output path: scenarios[0].networkResult.limitations[2] | Value: No motor/IBR dynamic or
generator subtransient decrement; no short-circuit peak, making/breaking duty
Calculated output path: scenarios[0].networkResult.limitations[3] | Value: No IEC 60909 voltage
correction by equipment type, maximum/minimum operating envelope or breaker rating
Calculated output path: scenarios[0].networkResult.limitations[4] | Value: No harmonic, nonlinear
electric arc, fault resistance estimation or conductor thermal qualification
Calculated output path: scenarios[1].id | Value: example_low
Calculated output path: scenarios[1].sourceReference | Value: illustrative-source-conditions-B
Calculated output path: scenarios[1].maximumPhaseRmsA | Value: 13555.382764706039
Calculated output path: scenarios[1].groundResidualA | Value: 13555.382764706039
Calculated output path: scenarios[1].faultType | Value: AG
Calculated output path: scenarios[1].networkResult.status | Value: BOUNDED_MULTI_BUS_SEQUENCE_FAULT
Calculated output path: scenarios[1].networkResult.scope | Value: One voltage base, explicitly
entered positive/negative/zero-sequence branch and source impedances, independent fixed
positive-sequence source EMFs; four metallic fault configurations. No IEC 60909 corrections,

transformer vector shifts, time-dependent generator/IBR currents or making/breaking duties.
Calculated output path: scenarios[1].networkResult.faultType | Value: AG
Calculated output path: scenarios[1].networkResult.nominalLLV | Value: 13800
Calculated output path: scenarios[1].networkResult.voltageFactor | Value: 0.95
Calculated output path: scenarios[1].networkResult.frequencyHz | Value: 60
Calculated output path: scenarios[1].networkResult.faultBus | Value: F
Calculated output path: scenarios[1].networkResult.faultImpedanceOhm.rOhm | Value: 0
Calculated output path: scenarios[1].networkResult.faultImpedanceOhm.xOhm | Value: 0
Calculated output path: scenarios[1].networkResult.sequenceTheveninOhm[0].real | Value:
0.2729322097008292
Calculated output path: scenarios[1].networkResult.sequenceTheveninOhm[0].imag | Value:
0.8628023435184697
Calculated output path: scenarios[1].networkResult.sequenceTheveninOhm[0].magnitude | Value:
0.9049419180660937
Calculated output path: scenarios[1].networkResult.sequenceTheveninOhm[0].angleDegrees | Value:
72.44615709955977
Calculated output path: scenarios[1].networkResult.sequenceTheveninOhm[1].real | Value:
0.07332704773129058
Calculated output path: scenarios[1].networkResult.sequenceTheveninOhm[1].imag | Value:
0.3106022392457279
Calculated output path: scenarios[1].networkResult.sequenceTheveninOhm[1].magnitude | Value:
0.31914041886518757
Calculated output path: scenarios[1].networkResult.sequenceTheveninOhm[1].angleDegrees | Value:
76.7168095145296
Calculated output path: scenarios[1].networkResult.sequenceTheveninOhm[2].real | Value:
0.08554316130525336
Calculated output path: scenarios[1].networkResult.sequenceTheveninOhm[2].imag | Value:
0.34845082156908147
Calculated output path: scenarios[1].networkResult.sequenceTheveninOhm[2].magnitude | Value:
0.3587974463374349
Calculated output path: scenarios[1].networkResult.sequenceTheveninOhm[2].angleDegrees | Value:
76.20691177731727
Calculated output path: scenarios[1].networkResult.prefaultPositiveFaultBusV.real | Value:
7147.8707117338045
Calculated output path: scenarios[1].networkResult.prefaultPositiveFaultBusV.imag | Value:
12.29247787397336
Calculated output path: scenarios[1].networkResult.prefaultPositiveFaultBusV.magnitude | Value:
7147.8812816578165
Calculated output path: scenarios[1].networkResult.prefaultPositiveFaultBusV.angleDegrees | Value:
0.09853373627072966
Calculated output path: scenarios[1].networkResult.sequenceFaultCurrentA[0].real | Value:
1240.8306532296388
Calculated output path: scenarios[1].networkResult.sequenceFaultCurrentA[0].imag | Value:
-4344.747218164595
Calculated output path: scenarios[1].networkResult.sequenceFaultCurrentA[0].magnitude | Value:
4518.46092156868
Calculated output path: scenarios[1].networkResult.sequenceFaultCurrentA[0].angleDegrees | Value:
-74.0610136249277
Calculated output path: scenarios[1].networkResult.sequenceFaultCurrentA[1].real | Value:
1240.8306532296388
Calculated output path: scenarios[1].networkResult.sequenceFaultCurrentA[1].imag | Value:
-4344.747218164595
Calculated output path: scenarios[1].networkResult.sequenceFaultCurrentA[1].magnitude | Value:
4518.46092156868
Calculated output path: scenarios[1].networkResult.sequenceFaultCurrentA[1].angleDegrees | Value:
-74.0610136249277
Calculated output path: scenarios[1].networkResult.sequenceFaultCurrentA[2].real | Value:
1240.8306532296388
Calculated output path: scenarios[1].networkResult.sequenceFaultCurrentA[2].imag | Value:

-4344.747218164595
Calculated output path: scenarios[1].networkResult.sequenceFaultCurrentA[2].magnitude | Value:
4518.46092156868
Calculated output path: scenarios[1].networkResult.sequenceFaultCurrentA[2].angleDegrees | Value:
-74.0610136249277
Calculated output path: scenarios[1].networkResult.phaseFaultCurrentA[0].real | Value:
3722.4919596889163
Calculated output path: scenarios[1].networkResult.phaseFaultCurrentA[0].imag | Value:
-13034.241654493786
Calculated output path: scenarios[1].networkResult.phaseFaultCurrentA[0].magnitude | Value:
13555.382764706039
Calculated output path: scenarios[1].networkResult.phaseFaultCurrentA[0].angleDegrees | Value:
-74.0610136249277
Calculated output path: scenarios[1].networkResult.phaseFaultCurrentA[1].real | Value:
4.547473508864641e-13
Calculated output path: scenarios[1].networkResult.phaseFaultCurrentA[1].imag | Value: 0
Calculated output path: scenarios[1].networkResult.phaseFaultCurrentA[1].magnitude | Value:
4.547473508864641e-13
Calculated output path: scenarios[1].networkResult.phaseFaultCurrentA[1].angleDegrees | Value: 0
Calculated output path: scenarios[1].networkResult.phaseFaultCurrentA[2].real | Value: 0
Calculated output path: scenarios[1].networkResult.phaseFaultCurrentA[2].imag | Value: 0
Calculated output path: scenarios[1].networkResult.phaseFaultCurrentA[2].magnitude | Value: 0
Calculated output path: scenarios[1].networkResult.phaseFaultCurrentA[2].angleDegrees | Value: 0
Calculated output path: scenarios[1].networkResult.groundResidualA | Value: 13555.382764706039
Calculated output path: scenarios[1].networkResult.maximumPhaseRmsA | Value: 13555.382764706039
Calculated output path: scenarios[1].networkResult.sourceContributions[0].id | Value: grid-A
Calculated output path: scenarios[1].networkResult.sourceContributions[0].bus | Value: A
Calculated output path: scenarios[1].networkResult.sourceContributions[0].phaseCurrentsA[0].real |
Value: 2375.779092476766
Calculated output path: scenarios[1].networkResult.sourceContributions[0].phaseCurrentsA[0].imag |
Value: -7970.9080996116645
Calculated output path: scenarios[1].networkResult.sourceContributions[0].phaseCurrentsA[0].magnitu
de | Value: 8317.433632359476
Calculated output path: scenarios[1].networkResult.sourceContributions[0].phaseCurrentsA[0].angleDe
grees | Value: -73.40298267951442
Calculated output path: scenarios[1].networkResult.sourceContributions[0].phaseCurrentsA[1].real |
Value: -787.4829867428919
Calculated output path: scenarios[1].networkResult.sourceContributions[0].phaseCurrentsA[1].imag |
Value: 438.1365747993191
Calculated output path: scenarios[1].networkResult.sourceContributions[0].phaseCurrentsA[1].magnitu
de | Value: 901.162090073914
Calculated output path: scenarios[1].networkResult.sourceContributions[0].phaseCurrentsA[1].angleDe
grees | Value: 150.90949050414696
Calculated output path: scenarios[1].networkResult.sourceContributions[0].phaseCurrentsA[2].real |
Value: 442.60710817447307
Calculated output path: scenarios[1].networkResult.sourceContributions[0].phaseCurrentsA[2].imag |
Value: 724.6773381512139
Calculated output path: scenarios[1].networkResult.sourceContributions[0].phaseCurrentsA[2].magnitu
de | Value: 849.1515157123013
Calculated output path: scenarios[1].networkResult.sourceContributions[0].phaseCurrentsA[2].angleDe
grees | Value: 58.58490084239294
Calculated output path: scenarios[1].networkResult.sourceContributions[0].sequenceCurrentsA[0].real
| Value: 676.9677379694488
Calculated output path: scenarios[1].networkResult.sourceContributions[0].sequenceCurrentsA[0].imag
| Value: -2269.364728887044
Calculated output path: scenarios[1].networkResult.sourceContributions[0].sequenceCurrentsA[0].magn
itude | Value: 2368.1852948973474
Calculated output path: scenarios[1].networkResult.sourceContributions[0].sequenceCurrentsA[0].angl
eDegrees | Value: -73.38978141401347

Calculated output path: scenarios[1].networkResult.sourceContributions[0].sequenceCurrentsA[1].real | Value: 932.1228706811672
Calculated output path: scenarios[1].networkResult.sourceContributions[0].sequenceCurrentsA[1].imag | Value: -3205.8681090763266
Calculated output path: scenarios[1].networkResult.sourceContributions[0].sequenceCurrentsA[1].magnitude | Value: 3338.628966932313
Calculated output path: scenarios[1].networkResult.sourceContributions[0].sequenceCurrentsA[1].angleDegrees | Value: -73.78793518777795
Calculated output path: scenarios[1].networkResult.sourceContributions[0].sequenceCurrentsA[2].real | Value: 766.6884838261499
Calculated output path: scenarios[1].networkResult.sourceContributions[0].sequenceCurrentsA[2].imag | Value: -2495.675261648294
Calculated output path: scenarios[1].networkResult.sourceContributions[0].sequenceCurrentsA[2].magnitude | Value: 2610.786518050628
Calculated output path: scenarios[1].networkResult.sourceContributions[0].sequenceCurrentsA[2].angleDegrees | Value: -72.92268466947286
Calculated output path: scenarios[1].networkResult.sourceContributions[1].id | Value: source-B
Calculated output path: scenarios[1].networkResult.sourceContributions[1].bus | Value: B
Calculated output path: scenarios[1].networkResult.sourceContributions[1].phaseCurrentsA[0].real | Value: 1346.712867212152
Calculated output path: scenarios[1].networkResult.sourceContributions[1].phaseCurrentsA[0].imag | Value: -5063.333554882115
Calculated output path: scenarios[1].networkResult.sourceContributions[1].phaseCurrentsA[0].magnitude | Value: 5239.3684957931655
Calculated output path: scenarios[1].networkResult.sourceContributions[1].phaseCurrentsA[0].angleDegrees | Value: -75.10566464442842
Calculated output path: scenarios[1].networkResult.sourceContributions[1].phaseCurrentsA[1].real | Value: 787.4829867429037
Calculated output path: scenarios[1].networkResult.sourceContributions[1].phaseCurrentsA[1].imag | Value: -438.1365747993241
Calculated output path: scenarios[1].networkResult.sourceContributions[1].phaseCurrentsA[1].magnitude | Value: 901.1620900739267
Calculated output path: scenarios[1].networkResult.sourceContributions[1].phaseCurrentsA[1].angleDegrees | Value: -29.090509495852952
Calculated output path: scenarios[1].networkResult.sourceContributions[1].phaseCurrentsA[2].real | Value: -442.6071081744835
Calculated output path: scenarios[1].networkResult.sourceContributions[1].phaseCurrentsA[2].imag | Value: -724.6773381512237
Calculated output path: scenarios[1].networkResult.sourceContributions[1].phaseCurrentsA[2].magnitude | Value: 849.1515157123149
Calculated output path: scenarios[1].networkResult.sourceContributions[1].phaseCurrentsA[2].angleDegrees | Value: -121.41509915760734
Calculated output path: scenarios[1].networkResult.sourceContributions[1].sequenceCurrentsA[0].real | Value: 563.8629152601907
Calculated output path: scenarios[1].networkResult.sourceContributions[1].sequenceCurrentsA[0].imag | Value: -2075.382489277554
Calculated output path: scenarios[1].networkResult.sourceContributions[1].sequenceCurrentsA[0].magnitude | Value: 2150.617089117823
Calculated output path: scenarios[1].networkResult.sourceContributions[1].sequenceCurrentsA[0].angleDegrees | Value: -74.80015495744995
Calculated output path: scenarios[1].networkResult.sourceContributions[1].sequenceCurrentsA[1].real | Value: 308.7077825484706
Calculated output path: scenarios[1].networkResult.sourceContributions[1].sequenceCurrentsA[1].imag | Value: -1138.8791090882573
Calculated output path: scenarios[1].networkResult.sourceContributions[1].sequenceCurrentsA[1].magnitude | Value: 1179.9771693230578
Calculated output path: scenarios[1].networkResult.sourceContributions[1].sequenceCurrentsA[1].angleDegrees | Value: -74.83368261136107
Calculated output path: scenarios[1].networkResult.sourceContributions[1].sequenceCurrentsA[2].real

| Value: 474.14216940349087
Calculated output path: scenarios[1].networkResult.sourceContributions[1].sequenceCurrentsA[2].imag
| Value: -1849.0719565163033
Calculated output path: scenarios[1].networkResult.sourceContributions[1].sequenceCurrentsA[2].magn
itude | Value: 1908.8944175049803
Calculated output path: scenarios[1].networkResult.sourceContributions[1].sequenceCurrentsA[2].angl
eDegrees | Value: -75.61799033819408
Calculated output path: scenarios[1].networkResult.branchCurrents[0].id | Value: line-A
Calculated output path: scenarios[1].networkResult.branchCurrents[0].from | Value: A
Calculated output path: scenarios[1].networkResult.branchCurrents[0].to | Value: F
Calculated output path: scenarios[1].networkResult.branchCurrents[0].phaseCurrentsA[0].real |
Value: 2375.7790924767637
Calculated output path: scenarios[1].networkResult.branchCurrents[0].phaseCurrentsA[0].imag |
Value: -7970.908099611659
Calculated output path: scenarios[1].networkResult.branchCurrents[0].phaseCurrentsA[0].magnitude |
Value: 8317.43363235947
Calculated output path: scenarios[1].networkResult.branchCurrents[0].phaseCurrentsA[0].angleDegrees
| Value: -73.40298267951442
Calculated output path: scenarios[1].networkResult.branchCurrents[0].phaseCurrentsA[1].real |
Value: -787.4829867428873
Calculated output path: scenarios[1].networkResult.branchCurrents[0].phaseCurrentsA[1].imag |
Value: 438.13657479931953
Calculated output path: scenarios[1].networkResult.branchCurrents[0].phaseCurrentsA[1].magnitude |
Value: 901.1620900739102
Calculated output path: scenarios[1].networkResult.branchCurrents[0].phaseCurrentsA[1].angleDegrees
| Value: 150.9094905041468
Calculated output path: scenarios[1].networkResult.branchCurrents[0].phaseCurrentsA[2].real |
Value: 442.607108174469
Calculated output path: scenarios[1].networkResult.branchCurrents[0].phaseCurrentsA[2].imag |
Value: 724.6773381512106
Calculated output path: scenarios[1].networkResult.branchCurrents[0].phaseCurrentsA[2].magnitude |
Value: 849.1515157122963
Calculated output path: scenarios[1].networkResult.branchCurrents[0].phaseCurrentsA[2].angleDegrees
| Value: 58.58490084239307
Calculated output path: scenarios[1].networkResult.branchCurrents[0].sequenceCurrentsA[0].real |
Value: 676.9677379694485
Calculated output path: scenarios[1].networkResult.branchCurrents[0].sequenceCurrentsA[0].imag |
Value: -2269.364728887043
Calculated output path: scenarios[1].networkResult.branchCurrents[0].sequenceCurrentsA[0].magnitude
| Value: 2368.1852948973465
Calculated output path: scenarios[1].networkResult.branchCurrents[0].sequenceCurrentsA[0].angleDegr
ees | Value: -73.38978141401347
Calculated output path: scenarios[1].networkResult.branchCurrents[0].sequenceCurrentsA[1].real |
Value: 932.1228706811652
Calculated output path: scenarios[1].networkResult.branchCurrents[0].sequenceCurrentsA[1].imag |
Value: -3205.868109076322
Calculated output path: scenarios[1].networkResult.branchCurrents[0].sequenceCurrentsA[1].magnitude
| Value: 3338.6289669323082
Calculated output path: scenarios[1].networkResult.branchCurrents[0].sequenceCurrentsA[1].angleDegr
ees | Value: -73.78793518777798
Calculated output path: scenarios[1].networkResult.branchCurrents[0].sequenceCurrentsA[2].real |
Value: 766.68848382615
Calculated output path: scenarios[1].networkResult.branchCurrents[0].sequenceCurrentsA[2].imag |
Value: -2495.675261648294
Calculated output path: scenarios[1].networkResult.branchCurrents[0].sequenceCurrentsA[2].magnitude
| Value: 2610.786518050628
Calculated output path: scenarios[1].networkResult.branchCurrents[0].sequenceCurrentsA[2].angleDegr
ees | Value: -72.92268466947286
Calculated output path: scenarios[1].networkResult.branchCurrents[1].id | Value: line-B

Calculated output path: scenarios[1].networkResult.branchCurrents[1].from | Value: B
Calculated output path: scenarios[1].networkResult.branchCurrents[1].to | Value: F
Calculated output path: scenarios[1].networkResult.branchCurrents[1].phaseCurrentsA[0].real |
Value: 1346.7128672121537
Calculated output path: scenarios[1].networkResult.branchCurrents[1].phaseCurrentsA[0].imag |
Value: -5063.333554882129
Calculated output path: scenarios[1].networkResult.branchCurrents[1].phaseCurrentsA[0].magnitude |
Value: 5239.368495793178
Calculated output path: scenarios[1].networkResult.branchCurrents[1].phaseCurrentsA[0].angleDegrees
| Value: -75.10566464442844
Calculated output path: scenarios[1].networkResult.branchCurrents[1].phaseCurrentsA[1].real |
Value: 787.4829867428903
Calculated output path: scenarios[1].networkResult.branchCurrents[1].phaseCurrentsA[1].imag |
Value: -438.1365747993168
Calculated output path: scenarios[1].networkResult.branchCurrents[1].phaseCurrentsA[1].magnitude |
Value: 901.1620900739114
Calculated output path: scenarios[1].networkResult.branchCurrents[1].phaseCurrentsA[1].angleDegrees
| Value: -29.090509495852963
Calculated output path: scenarios[1].networkResult.branchCurrents[1].phaseCurrentsA[2].real |
Value: -442.60710817447193
Calculated output path: scenarios[1].networkResult.branchCurrents[1].phaseCurrentsA[2].imag |
Value: -724.6773381512149
Calculated output path: scenarios[1].networkResult.branchCurrents[1].phaseCurrentsA[2].magnitude |
Value: 849.1515157123015
Calculated output path: scenarios[1].networkResult.branchCurrents[1].phaseCurrentsA[2].angleDegrees
| Value: -121.41509915760699
Calculated output path: scenarios[1].networkResult.branchCurrents[1].sequenceCurrentsA[0].real |
Value: 563.8629152601907
Calculated output path: scenarios[1].networkResult.branchCurrents[1].sequenceCurrentsA[0].imag |
Value: -2075.3824892775533
Calculated output path: scenarios[1].networkResult.branchCurrents[1].sequenceCurrentsA[0].magnitude
| Value: 2150.6170891178217
Calculated output path: scenarios[1].networkResult.branchCurrents[1].sequenceCurrentsA[0].angleDegr
ees | Value: -74.80015495744995
Calculated output path: scenarios[1].networkResult.branchCurrents[1].sequenceCurrentsA[1].real |
Value: 308.7077825484718
Calculated output path: scenarios[1].networkResult.branchCurrents[1].sequenceCurrentsA[1].imag |
Value: -1138.8791090882717
Calculated output path: scenarios[1].networkResult.branchCurrents[1].sequenceCurrentsA[1].magnitude
| Value: 1179.9771693230718
Calculated output path: scenarios[1].networkResult.branchCurrents[1].sequenceCurrentsA[1].angleDegr
ees | Value: -74.8336826113612
Calculated output path: scenarios[1].networkResult.branchCurrents[1].sequenceCurrentsA[2].real |
Value: 474.14216940349115
Calculated output path: scenarios[1].networkResult.branchCurrents[1].sequenceCurrentsA[2].imag |
Value: -1849.071956516303
Calculated output path: scenarios[1].networkResult.branchCurrents[1].sequenceCurrentsA[2].magnitude
| Value: 1908.8944175049803
Calculated output path: scenarios[1].networkResult.branchCurrents[1].sequenceCurrentsA[2].angleDegr
ees | Value: -75.61799033819408
Calculated output path: scenarios[1].networkResult.busVoltages[0].bus | Value: F
Calculated output path: scenarios[1].networkResult.busVoltages[0].phaseVoltagesV[0].real | Value:
1.5916157281026244e-12
Calculated output path: scenarios[1].networkResult.busVoltages[0].phaseVoltagesV[0].imag | Value:
3.197442310920451e-13
Calculated output path: scenarios[1].networkResult.busVoltages[0].phaseVoltagesV[0].magnitude |
Value: 1.6234152270014871e-12
Calculated output path: scenarios[1].networkResult.busVoltages[0].phaseVoltagesV[0].angleDegrees |
Value: 11.359113388562097

Calculated output path: scenarios[1].networkResult.busVoltages[0].phaseVoltagesV[1].real | Value:
-6125.628861574908
Calculated output path: scenarios[1].networkResult.busVoltages[0].phaseVoltagesV[1].imag | Value:
-6172.931550590695
Calculated output path: scenarios[1].networkResult.busVoltages[0].phaseVoltagesV[1].magnitude |
Value: 8696.459789939672
Calculated output path: scenarios[1].networkResult.busVoltages[0].phaseVoltagesV[1].angleDegrees |
Value: -134.77963021808338
Calculated output path: scenarios[1].networkResult.busVoltages[0].phaseVoltagesV[2].real | Value:
-6136.333340059855
Calculated output path: scenarios[1].networkResult.busVoltages[0].phaseVoltagesV[2].imag | Value:
6518.621140578018
Calculated output path: scenarios[1].networkResult.busVoltages[0].phaseVoltagesV[2].magnitude |
Value: 8952.486159426375
Calculated output path: scenarios[1].networkResult.busVoltages[0].phaseVoltagesV[2].angleDegrees |
Value: 133.2697038943233
Calculated output path: scenarios[1].networkResult.busVoltages[0].sequenceVoltagesV[0].real |
Value: -4087.3207338782536
Calculated output path: scenarios[1].networkResult.busVoltages[0].sequenceVoltagesV[0].imag |
Value: 115.22986332910773
Calculated output path: scenarios[1].networkResult.busVoltages[0].sequenceVoltagesV[0].magnitude |
Value: 4088.944693071051
Calculated output path: scenarios[1].networkResult.busVoltages[0].sequenceVoltagesV[0].angleDegrees
| Value: 178.3851434746321
Calculated output path: scenarios[1].networkResult.busVoltages[0].sequenceVoltagesV[1].real |
Value: 5707.3960482794155
Calculated output path: scenarios[1].networkResult.busVoltages[0].sequenceVoltagesV[1].imag |
Value: -54.524814897144516
Calculated output path: scenarios[1].networkResult.busVoltages[0].sequenceVoltagesV[1].magnitude |
Value: 5707.656489957595
Calculated output path: scenarios[1].networkResult.busVoltages[0].sequenceVoltagesV[1].angleDegrees
| Value: -0.5473506150233923
Calculated output path: scenarios[1].networkResult.busVoltages[0].sequenceVoltagesV[2].real |
Value: -1620.0753144011603
Calculated output path: scenarios[1].networkResult.busVoltages[0].sequenceVoltagesV[2].imag |
Value: -60.705048431962894
Calculated output path: scenarios[1].networkResult.busVoltages[0].sequenceVoltagesV[2].magnitude |
Value: 1621.2122400343349
Calculated output path: scenarios[1].networkResult.busVoltages[0].sequenceVoltagesV[2].angleDegrees
| Value: -177.85410184761042
Calculated output path: scenarios[1].networkResult.busVoltages[1].bus | Value: A
Calculated output path: scenarios[1].networkResult.busVoltages[1].phaseVoltagesV[0].real | Value:
1997.9173362718939
Calculated output path: scenarios[1].networkResult.busVoltages[1].phaseVoltagesV[0].imag | Value:
-371.9463055800213
Calculated output path: scenarios[1].networkResult.busVoltages[1].phaseVoltagesV[0].magnitude |
Value: 2032.2445071423876
Calculated output path: scenarios[1].networkResult.busVoltages[1].phaseVoltagesV[0].angleDegrees |
Value: -10.545856900673154
Calculated output path: scenarios[1].networkResult.busVoltages[1].phaseVoltagesV[1].real | Value:
-5404.629448406164
Calculated output path: scenarios[1].networkResult.busVoltages[1].phaseVoltagesV[1].imag | Value:
-6355.626070240976
Calculated output path: scenarios[1].networkResult.busVoltages[1].phaseVoltagesV[1].magnitude |
Value: 8342.901307057746
Calculated output path: scenarios[1].networkResult.busVoltages[1].phaseVoltagesV[1].angleDegrees |
Value: -130.37681923288432
Calculated output path: scenarios[1].networkResult.busVoltages[1].phaseVoltagesV[2].real | Value:
-5392.580908170675

Calculated output path: scenarios[1].networkResult.busVoltages[1].phaseVoltagesV[2].imag | Value:
6400.997716995932
Calculated output path: scenarios[1].networkResult.busVoltages[1].phaseVoltagesV[2].magnitude |
Value: 8369.749137468458
Calculated output path: scenarios[1].networkResult.busVoltages[1].phaseVoltagesV[2].angleDegrees |
Value: 130.11277836379037
Calculated output path: scenarios[1].networkResult.busVoltages[1].sequenceVoltagesV[0].real |
Value: -2933.0976734349815
Calculated output path: scenarios[1].networkResult.busVoltages[1].sequenceVoltagesV[0].imag |
Value: -108.85821960835483
Calculated output path: scenarios[1].networkResult.busVoltages[1].sequenceVoltagesV[0].magnitude |
Value: 2935.1170460283183
Calculated output path: scenarios[1].networkResult.busVoltages[1].sequenceVoltagesV[0].angleDegrees
| Value: -177.87451497433642
Calculated output path: scenarios[1].networkResult.busVoltages[1].sequenceVoltagesV[1].real |
Value: 6148.027593609444
Calculated output path: scenarios[1].networkResult.busVoltages[1].sequenceVoltagesV[1].imag |
Value: -135.02215695998402
Calculated output path: scenarios[1].networkResult.busVoltages[1].sequenceVoltagesV[1].magnitude |
Value: 6149.510084116723
Calculated output path: scenarios[1].networkResult.busVoltages[1].sequenceVoltagesV[1].angleDegrees
| Value: -1.258119973086647
Calculated output path: scenarios[1].networkResult.busVoltages[1].sequenceVoltagesV[2].real |
Value: -1217.0125839025686
Calculated output path: scenarios[1].networkResult.busVoltages[1].sequenceVoltagesV[2].imag |
Value: -128.0659290116825
Calculated output path: scenarios[1].networkResult.busVoltages[1].sequenceVoltagesV[2].magnitude |
Value: 1223.7322058158115
Calculated output path: scenarios[1].networkResult.busVoltages[1].sequenceVoltagesV[2].angleDegrees
| Value: -173.99288724741223
Calculated output path: scenarios[1].networkResult.busVoltages[2].bus | Value: B
Calculated output path: scenarios[1].networkResult.busVoltages[2].phaseVoltagesV[0].real | Value:
1600.1946532066663
Calculated output path: scenarios[1].networkResult.busVoltages[2].phaseVoltagesV[0].imag | Value:
-207.943951669656
Calculated output path: scenarios[1].networkResult.busVoltages[2].phaseVoltagesV[0].magnitude |
Value: 1613.6491611212134
Calculated output path: scenarios[1].networkResult.busVoltages[2].phaseVoltagesV[0].angleDegrees |
Value: -7.4040476254726535
Calculated output path: scenarios[1].networkResult.busVoltages[2].phaseVoltagesV[1].real | Value:
-5235.704523615647
Calculated output path: scenarios[1].networkResult.busVoltages[2].phaseVoltagesV[1].imag | Value:
-6210.021335523312
Calculated output path: scenarios[1].networkResult.busVoltages[2].phaseVoltagesV[1].magnitude |
Value: 8122.620688316309
Calculated output path: scenarios[1].networkResult.busVoltages[2].phaseVoltagesV[1].angleDegrees |
Value: -130.13442954875794
Calculated output path: scenarios[1].networkResult.busVoltages[2].phaseVoltagesV[2].real | Value:
-5265.752022594378
Calculated output path: scenarios[1].networkResult.busVoltages[2].phaseVoltagesV[2].imag | Value:
6205.865032231779
Calculated output path: scenarios[1].networkResult.busVoltages[2].phaseVoltagesV[2].magnitude |
Value: 8138.851587400641
Calculated output path: scenarios[1].networkResult.busVoltages[2].phaseVoltagesV[2].angleDegrees |
Value: 130.31501040097731
Calculated output path: scenarios[1].networkResult.busVoltages[2].sequenceVoltagesV[0].real |
Value: -2967.087297667786
Calculated output path: scenarios[1].networkResult.busVoltages[2].sequenceVoltagesV[0].imag |
Value: -70.7000849870625

Calculated output path: scenarios[1].networkResult.busVoltages[2].sequenceVoltagesV[0].magnitude | Value: 2967.929502868743
Calculated output path: scenarios[1].networkResult.busVoltages[2].sequenceVoltagesV[0].angleDegrees | Value: -178.63500811410867
Calculated output path: scenarios[1].networkResult.busVoltages[2].sequenceVoltagesV[1].real | Value: 5867.79864376283
Calculated output path: scenarios[1].networkResult.busVoltages[2].sequenceVoltagesV[1].imag | Value: -59.94796752937404
Calculated output path: scenarios[1].networkResult.busVoltages[2].sequenceVoltagesV[1].magnitude | Value: 5868.104862948157
Calculated output path: scenarios[1].networkResult.busVoltages[2].sequenceVoltagesV[1].angleDegrees | Value: -0.5853380874235742
Calculated output path: scenarios[1].networkResult.busVoltages[2].sequenceVoltagesV[2].real | Value: -1300.5166928883773
Calculated output path: scenarios[1].networkResult.busVoltages[2].sequenceVoltagesV[2].imag | Value: -77.29589915321947
Calculated output path: scenarios[1].networkResult.busVoltages[2].sequenceVoltagesV[2].magnitude | Value: 1302.8116995587761
Calculated output path: scenarios[1].networkResult.busVoltages[2].sequenceVoltagesV[2].angleDegrees | Value: -176.5986403483676
Calculated output path: scenarios[1].networkResult.kclRelativeResidual | Value: 2.4237824915186094e-15
Calculated output path: scenarios[1].networkResult.nodalRelativeResidual | Value: 1.1443916996305582e-15
Calculated output path: scenarios[1].networkResult.limitations[0] | Value: Explicit positive/negative/zero-sequence equivalent impedances; no transformer winding shift or zero-sequence grounding topology inference
Calculated output path: scenarios[1].networkResult.limitations[1] | Value: Multiple documented fixed positive-sequence Thevenin EMFs, one voltage level only
Calculated output path: scenarios[1].networkResult.limitations[2] | Value: No motor/IBR dynamic or generator subtransient decrement; no short-circuit peak, making/breaking duty
Calculated output path: scenarios[1].networkResult.limitations[3] | Value: No IEC 60909 voltage correction by equipment type, maximum/minimum operating envelope or breaker rating
Calculated output path: scenarios[1].networkResult.limitations[4] | Value: No harmonic, nonlinear electric arc, fault resistance estimation or conductor thermal qualification
Calculated output path: limits[0] | Value: These are extrema of the entered cases ONLY. No automatic utility maximum/minimum correction regimes or operating-state enumeration.
Calculated output path: limits[1] | Value: No time-dependence or IBR/motor model; no breaker-making/breaking duty
Calculated output path: limits[2] | Value: All scenario impedances, source EMF angles, grounding paths and statuses must be documented; individual studies contain source contributions and KCL checks

===== ASSUMPTIONS & QA =====
Evidence / limitation: Release status | Value: BOUNDED METHOD, NOT A STANDARDS/DEVICE APPROVAL
Evidence / limitation: Warning | Value: ENGINEERING SCREENING ONLY - independently verify actual manufacturer, utility, installation, equipment settings and standards edition before professional use.
Evidence / limitation: Study method | Value: multienvelope
Evidence / limitation: Timestamp | Value: 2026-10-26T06:04:37.512Z